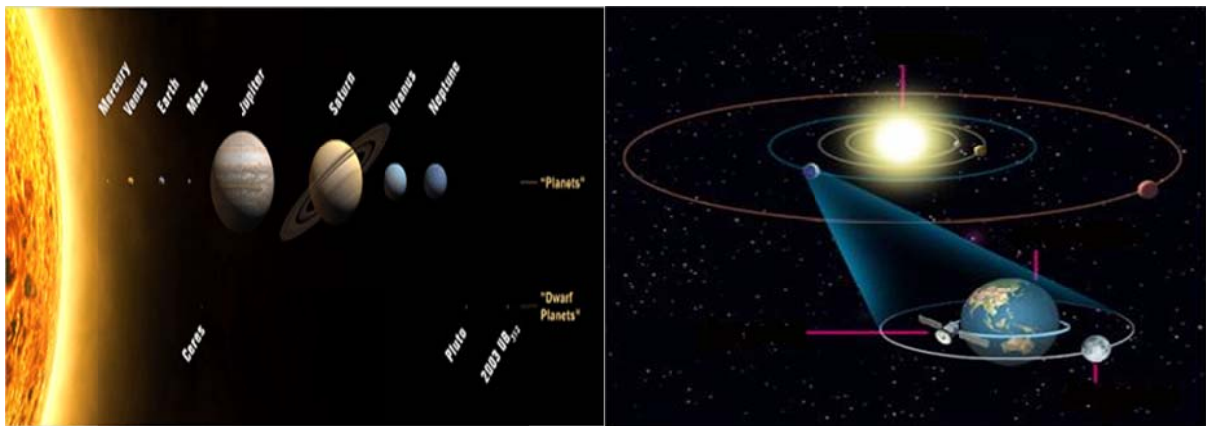


Chapter 1. Equations of Motion

- Definition
 - **Astrodynamics** studies the motion of man-made objects in space, subject to both natural and artificially induced forces
 - **Celestial Mechanics** examines the dynamic motions of celestial objects
 - **Orbital Dynamics** studies the motion of all orbiting bodies
 - * **Astrodynamics** combines feature of celestial mechanics with orbital dynamics



1.1. History

- Ptolemy
 - The Earth is the center of the Universe
 - Excentric and Epicycle Systems

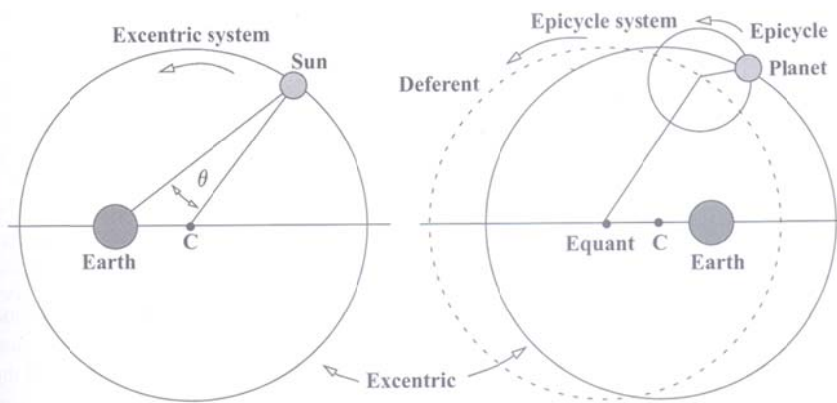


Figure 1-1. Excentric and Epicycle Systems. Both the excentric and epicycle theories adhered to an Earth-centered philosophy and used combinations of circular motion to approximate observations. The excentric system consisted of the Sun traveling around a larger circle, the excentric, with the Earth displaced to one side. In the epicycle system the Sun (or planet) moved around a smaller circle, the epicycle, the center of which moved around the equant, forming the deferent. Ptolemy placed the Earth at the equant for the Sun, but left it as shown for all other planets. (Beer and Strand, 1975b:559–560).

- Nicholas Copernicus (1473-1543)
 - The first scientist to bridge the gap between antiquity and modern times
 - Different from Ptolemaic theory in three main ways
 - I. Heliocentric
 - II. New numbers and data
 - III. Details of planetary motion



Nicolaus Copernicus, 1473–1543

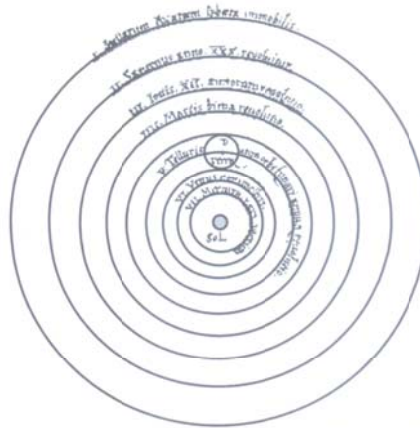
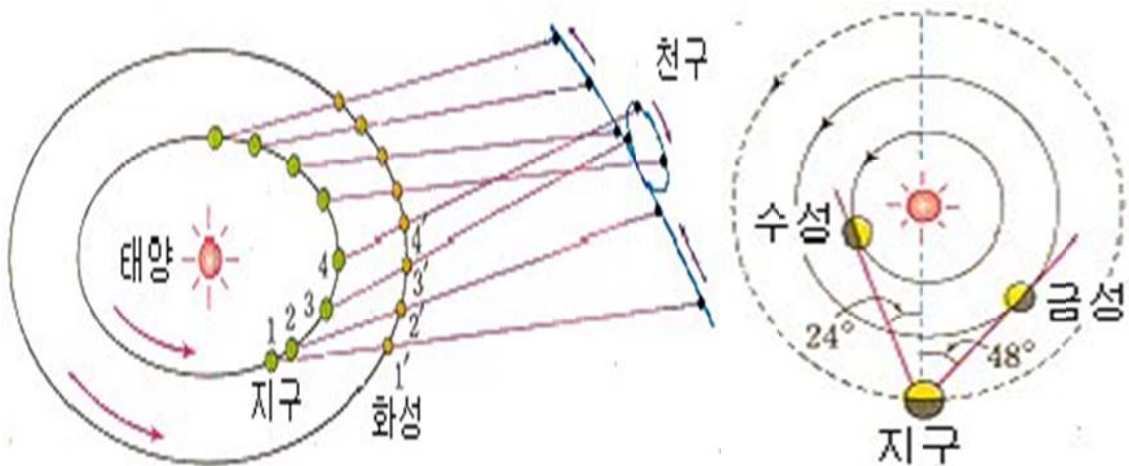


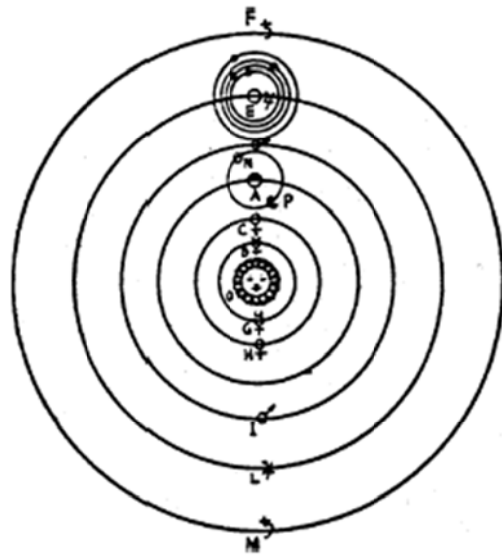
Figure 1-2. Copernicus's Solar System. (Pannekoek, 1989:191) In addition to defining a Sun-centered system, Copernicus also introduced smaller epicycles than the previous Ptolemaic system.



- Galileo Galilei (1564-1642)
 - Picked up where Copernicus left off
 - Used telescope for regular and dedicated scientific research
 - Provided foundation for the continuing work by Tycho Brahe, Johann Kepler, and Isaac Newton



Galileo Galilei, 1546–1642



- Johann Kepler (1571-1630)

- Took advantage of Tycho Brahe's vast and accurate observational data
- 'New Astronomy' (1609) contains his first two laws
- 'Harmony of the World' (1619) contains his third law.
- Kepler's Three Laws

- I. **Elliptic Motion:** The orbit of each planet is an ellipse with the Sun at one focus.
 - II. **Constant Aerial Speed:** The line joining the planet to the Sun sweeps out equal areas in equal times.
 - III. **Harmony:** The square of the period of a planet is proportional to the cube of its mean distance to the Sun.
- * Though remarkable, these laws did not completely solve planetary motion, but only captured the kinematics of motion

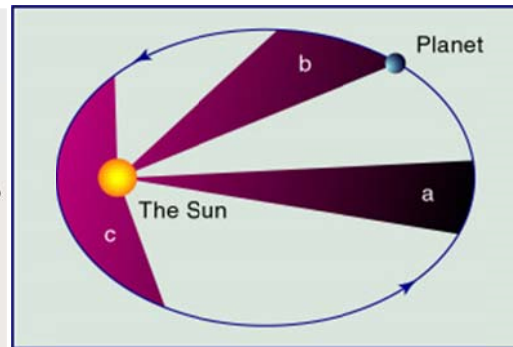
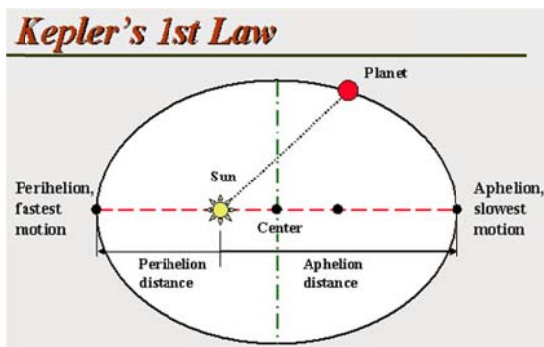
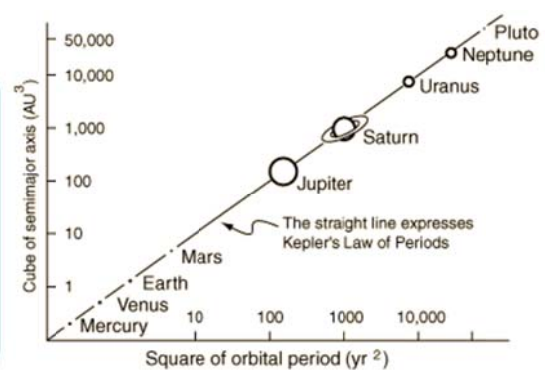


TABLE 2-3 A Demonstration of Kepler's Third Law

	Sidereal period P (yr)	Semimajor axis a (AU)	P^2	a^3
Mercury	0.24	0.39	0.06	0.06
Venus	0.61	0.72	0.37	0.37
Earth	1.00	1.00	1.00	1.00
Mars	1.88	1.52	3.53	3.51
Jupiter	11.86	5.20	140.7	140.6
Saturn	29.46	9.54	867.9	868.3
Uranus	84.01	19.19	7,058	7,067
Neptune	164.79	30.06	27,160	27,160
Pluto	248.54	39.53	61,770	61,770



- Isaac Newton (1642-1727)
- Unlocked the dynamics of motion

- 'Mathematical Principles of Natural Philosophy' or simply 'Principia' (1686)
- Newton's three laws of motion:
 - I. **Inertia:** everybody continues in its state of rest, or of uniform motion in a right (straight) line unless it is compelled to change that state by forces impressed upon it
 - II. $\vec{F} = m\vec{a}$: The change of motion is proportional to the motive force impressed and is made in the direction of the right line in which force is impressed
 - III. **Action-Reaction:** To every action there is always an opposed equal reaction, or the mutual actions of two bodies upon each other are always equal and directed to contrary parts
- ◆ The first law is simple consequence of the second law, if $\vec{F} = 0$
- ◆ The acceleration must be calculated with respect to an inertial frame of reference
- ◆ Absolute rest cannot be defined so that we conveniently assume inertial frame with respect to the representation of motion
- ◆ 'Inertial enough' depends on the problem at hand and on the accuracy requirements for the system



Isaac Newton, 1642–1727

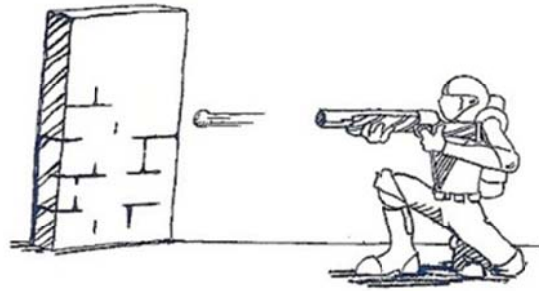


Figure 4-9. Newton's First Law. Any object in motion, such as a speeding bullet, will tend to stay in motion, in a straight line, unless acted on by some outside force (like gravity or hitting a brick wall.)

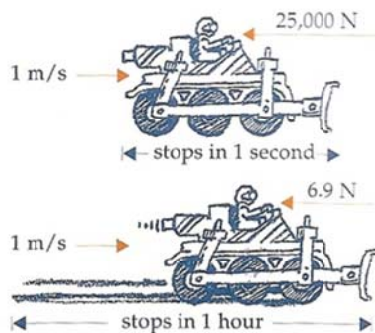


Figure 4-14. Newton's Second Law. The force we must apply to stop a moving object depends on how fast we want to change its momentum. If the two bulldozers are both moving at 1 m/s (about the speed of a brisk walk), you must apply a much, much larger force to stop a bulldozer in one second than to stop it in one hour.

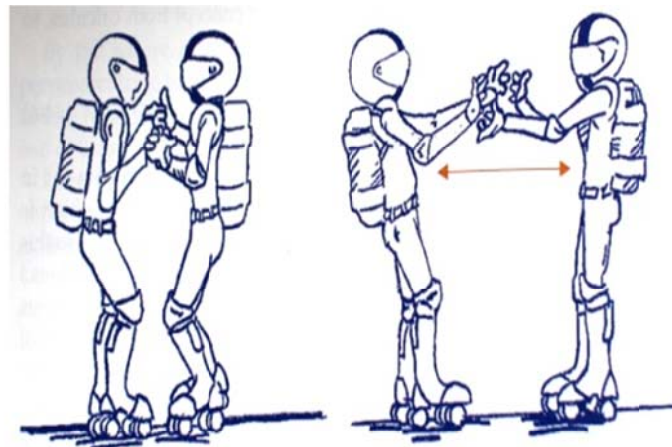


Figure 4-15. Two Roller Skaters Demonstrate Newton's Third Law. If they initially start at rest and the first one pushes against the second, they'll both go backwards. The first astronaut applied an action—pushing—and received an equal but opposite reaction.

■ Solution Procedure

- I. Select a reference frame
- II. Identify all the forces acting on the particle
- III. Set equations of motions by using the second law, $\vec{F} = m\vec{a}$
- IV. Solve the equations of motion

1.2. Geometry of Conic Sections

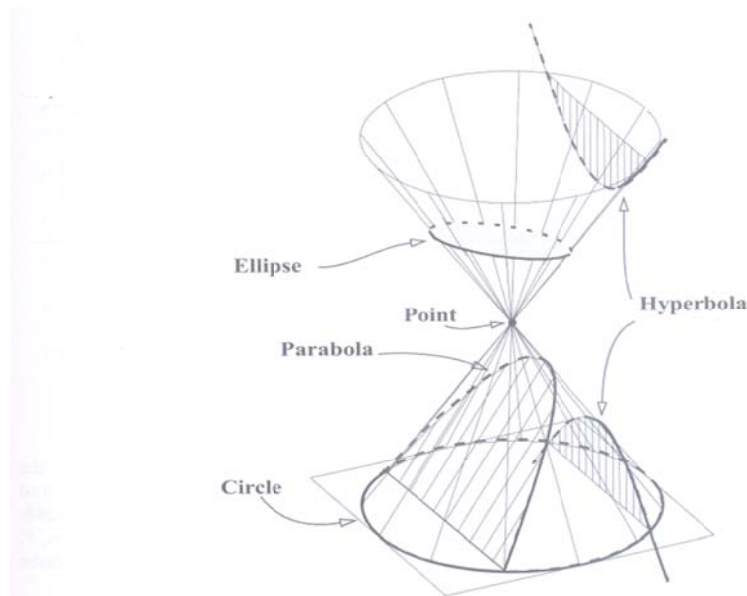


Figure 1-3. Conic Sections. Slicing a cone with a plane forms a conic section. When the plane is perpendicular to the axis of revolution, a circle results. Planes that are parallel to the axis of revolution yield hyperbolas (a pair as shown), and planes parallel to the outer surface yield parabolas. All other sections are ellipses, except for special cases in which the plane is *on* the surface (a line or rectilinear orbit) or only through the vertex (a point).

- Kepler's first law: Planets travel in ellipses, which are a part of conic sections
- Conic Section: Intersection of a plane and a right circular cone
 - Every conic section has two foci
 - Geometrically, the foci provide the locations from which to construct a conic section as illustrated in Fig. 1-4
 - In Astrodynamics, the gravitational center of attraction coincides with one focus for all orbital motion, called the *primary focus*

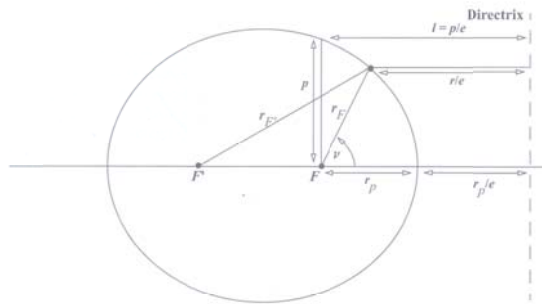


Figure 1-4. Geometry for Conic Sections. We create conic sections by recognizing that the sum of the distance from both foci to *any* point on the orbit is constant. The ratio of the distance from a focus to the orbit and the distance to the directrix is also a constant called the eccentricity, e . The closest point in the orbit to the primary focus, F , is the radius of periapsis, r_P . The distance l is a standard quantity used to describe conic sections.

- Figure 1-5: An Ellipse has two distinct foci (Primary & Secondary Focus)
- Figure 1-6: A circle is a special case in which both foci coincide
- Figure 1-7: A Parabola has one focus at infinity
- Figure 1-8: A Hyperbola has a branch associated with each focus

- Open Orbits: Parabolic and Hyperbolic Orbits do not repeat their position
- Closed Orbits: Satellites in Elliptic Orbits ideally retrace their positions over time

- Alternative Definition
 - Ellipse: Set of all points where the sum of the lengths from two points (foci) is constant

$$r_{F'} + r_F = \text{constant} = 2a, \quad a: \text{semimajor axis}$$

- Hyperbola: Set of all points where the difference between the lengths from two points (foci) is constant

$$r_{F'} - r_F = \text{constant} = -2a, \quad a: \text{semimajor axis}$$

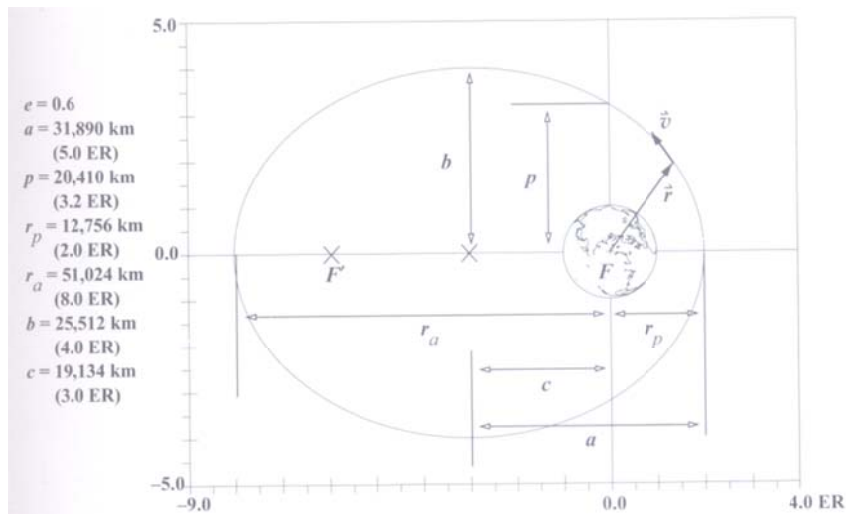


Figure 1-5. Elliptical Orbits. An elliptical orbit has two distinct foci with the primary focus, F , at the center of the Earth (or central body). The radius of apoapsis, r_a , and periapsis, r_p , denote the extreme points of the ellipse. The semimajor axis, a , and the semiminor axis, b , describe the shape of the orbit. Half the distance between the foci is c , and the semiparameter, p , locates the orbit distance normal from the semimajor axis at the focus. One ER is equal to the Earth's radius.

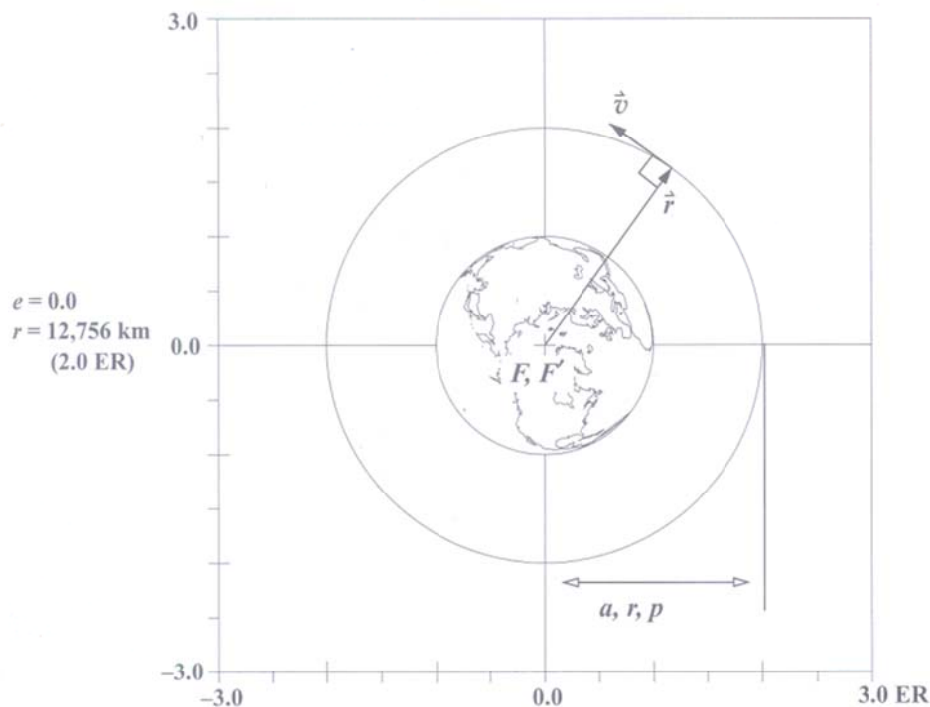


Figure 1-6. Circular Orbits. For circular orbits, the semimajor axis, semiparameter, and radius are always equal and constant. This is a special case of the ellipse, in which both foci coincide.

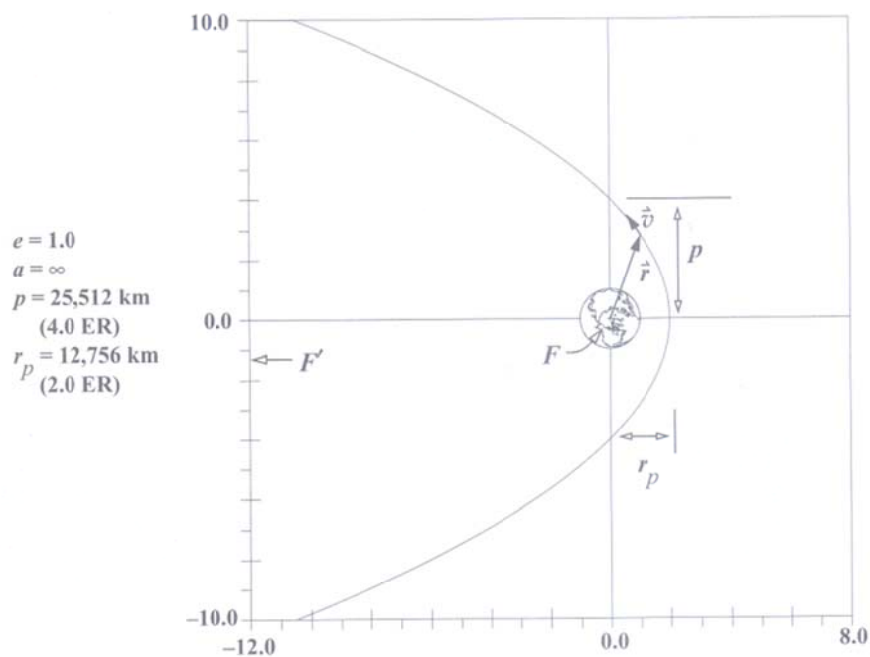


Figure 1-7. Parabolic Orbits. Parabolic orbits are considered open because they don't repeat. The semimajor axis is infinite. The second focus is at infinity.

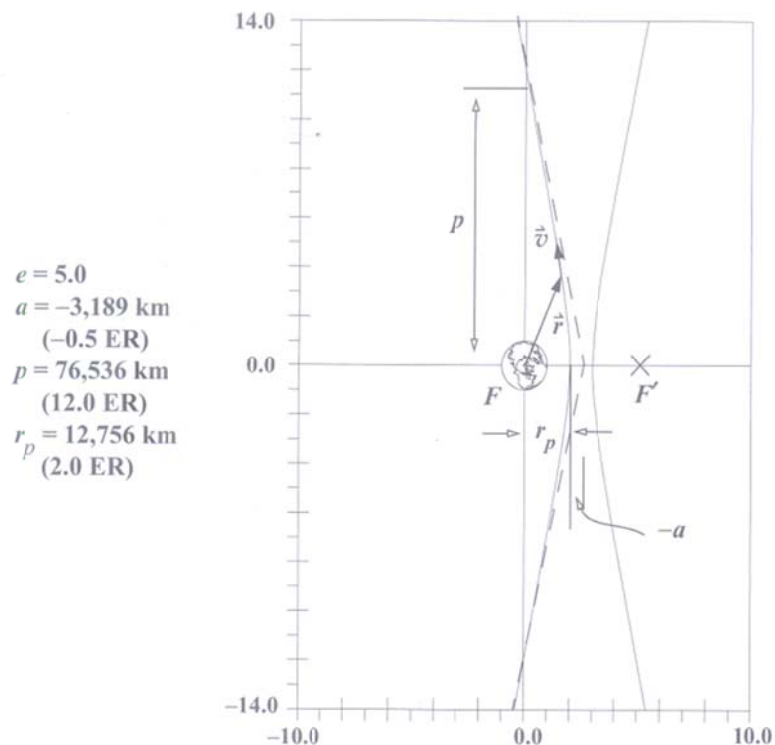


Figure 1-8. Hyperbolic Orbits. For a hyperbolic orbit, the semimajor axis is negative, and the eccentricity is larger than unity. Notice the rapid departure from the Earth with a modest eccentricity. The orbit approaches the dashed-line asymptotes. Hyperbolas occur in pairs called branches. The second branch is a mirror image of the first branch about the conjugate axis, shown as the dashed vertical line.

- Geometrical Parameters

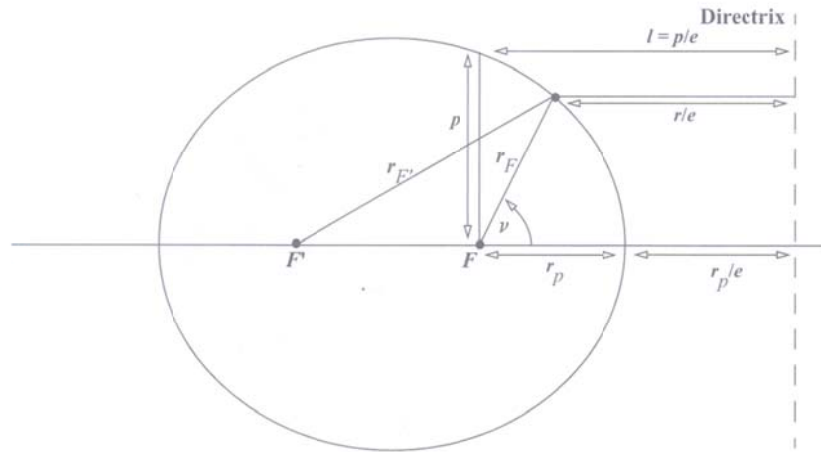


Figure 1-4. **Geometry for Conic Sections.** We create conic sections by recognizing that the sum of the distance from both foci to *any* point on the orbit is constant. The ratio of the distance from a focus to the orbit and the distance to the directrix is also a constant called the eccentricity, e . The closest point in the orbit to the primary focus, F , is the radius of periapsis, r_p . The distance l is a standard quantity used to describe conic sections.

- Semimajor axis (a)

- ◆ Scales the conic section

I. $a > 0$ for circles and ellipses

II. $a \rightarrow \infty$ for parabolas

III. $a < 0$ for hyperbolas

- ◆ Alternatively use semiminor axis (b) or half the distance between the foci (c)

- Eccentricity (e)

- ◆ Constant ratio of the distance from either focus to the orbit and the distance to the Directrix (a stationary line)

- ◆ Indicator of 'roundness' or 'flatness'

- ◆ Measure of departure from a circle
- ◆ The value determines the type of conic section for common orbits

- $e = 0$ for circles
- $e \in (0, 1)$ ellipses
- $e = 1$ parabolic and rectilinear orbits
- $e > 1$ hyperbolas

➤ Alternatively, Flattening (f) may be defined to describe the shape: $f = \frac{a-b}{a}$

■ Semiparameter; Semilatus Rectum; Parameter (p)

- ◆ Vertical distance of an orbit from the foci

■ Apoapsis (Farthest Point, r_a) ↔ Periapsis (Closest Point, r_p)

- ◆ Apogee-Perigee (Earth)
- ◆ Aphelion-Perihelion (Sun)
- ◆ Aposelenium-Periselenium (Moon)

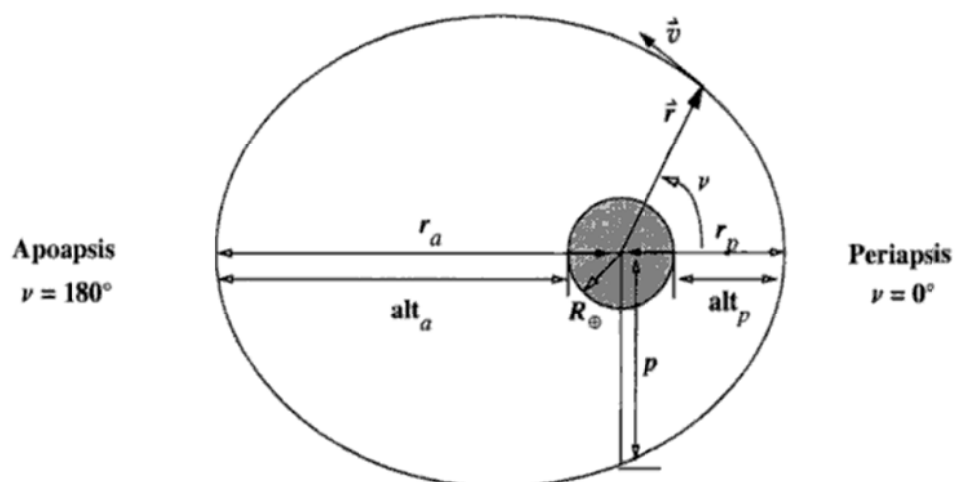


Figure 1-9. Radius of Apoapsis and Periapsis, as well as True Anomaly. Apoapsis occurs farthest from the planet; periapsis is closest. Notice the difference between altitudes and radius values. True anomaly, ν , locates the satellite's current position. We measure it in the direction of satellite motion from periapsis.

- True Anomaly (ν)

- ◆ Locates the satellite in the orbital plane
- ◆ Angular displacement measured from the periapsis to the position vector along the direction of motion

- Flight-Path Angle(ϕ_{fpa})

- ◆ Angle from the local horizontal (the line perpendicular to the radius vector) to the velocity vector
- $\phi_{fpa} > 0$ for $0^\circ < \nu < 180^\circ$
- $\phi_{fpa} < 0$ for $180^\circ < \nu < 360^\circ$

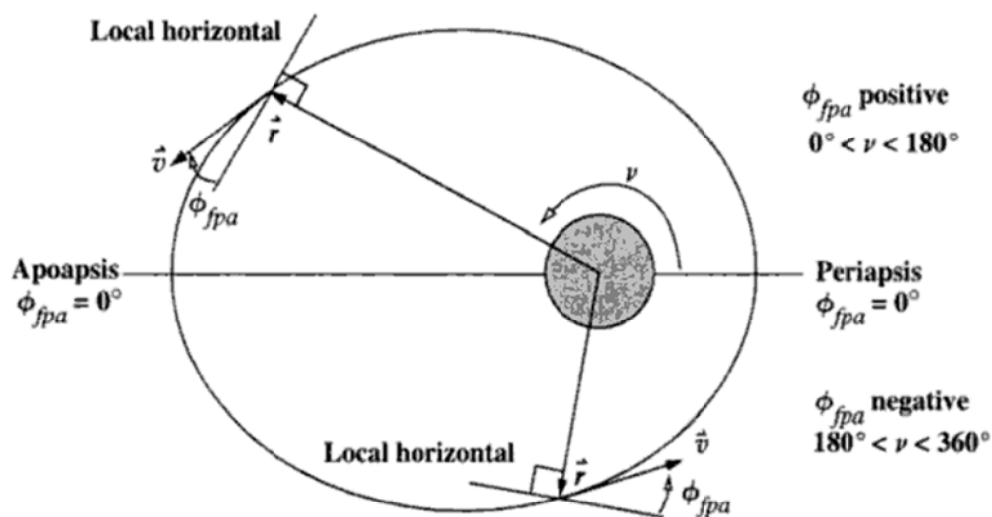


Figure 1-10. Geometry for the Flight-path Angle. The flight-path angle is always measured from the local horizontal to the velocity vector. It's always positive while the satellite travels from periapsis to apoapsis and negative for travel from apoapsis to periapsis. I've exaggerated the diagram for clarity.

- Mathematical Relations between Geometrical Parameters

- (Assumed to be given) From the definition of eccentricity, it turns out to be

$$e = \frac{c}{a}$$

Then from the geometry (refer to the diagram)

$$\begin{aligned} \frac{p}{e} &= \frac{r}{e} + r \cos \nu \\ \Rightarrow p &= r(1 + e \cos \nu) \\ \Rightarrow r &= \frac{p}{1 + e \cos \nu} \quad (\text{Trajectory Equation}) \end{aligned}$$

$$a^2 = b^2 + c^2$$

$$\begin{aligned} b^2 &= a^2 - c^2 \\ &= a^2 - a^2 e^2 \quad \leftarrow e = \frac{c}{a} \\ \Rightarrow b &= a\sqrt{1 - e^2} \quad \text{or} \quad e = \frac{\sqrt{a^2 - b^2}}{a} \end{aligned}$$

$$\begin{aligned} a &= c + r_p \\ \Rightarrow &= ae + r_p \quad \leftarrow e = \frac{c}{a} \\ \Rightarrow r_p &= a(1 - e) \end{aligned}$$

$$\begin{aligned} a + c &= r_a \\ \Rightarrow a + ae &= r_a \quad \leftarrow c = ae \\ \Rightarrow r_a &= a(1 + e) \end{aligned}$$

- From the trajectory equation, at $r = r_p$

$$\begin{aligned} r_p &= \frac{p}{1 + e} \\ \Rightarrow p &= r_p(1 + e) \\ \Rightarrow p &= a(1 - e^2) \end{aligned}$$

- From IV, $b = a\sqrt{1 - e^2}$

Analogously, $y_e = y_c \sqrt{1 - e^2}$

This expression is useful because it relates any vertical part of an ellipse, y_e to the corresponding part of an auxiliary circle, y_c

- From IV and VII,

$$p = \frac{b^2}{a}$$

- Relation between Eccentricity and Flattening

$$\begin{aligned} f &= \frac{a-b}{a} \\ &= 1 - \frac{\sqrt{a^2 - c^2}}{a} \\ &= 1 - \sqrt{1 - e^2} \\ \Rightarrow 1 - e^2 &= (1 - f)^2 \end{aligned}$$

TABLE 1-1. Values for Conic Sections. This table shows characteristic values for the basic parameters. Sec. 2.2 discusses the limited values.

	a	e	p	ν	ϕ_{fpa}
Circle	$a = r$	$e = 0$	$p = r$	Undefined	$\phi_{fpa} = 0$
Ellipse	$r_p \leq a \leq r_a$	$0 < e < 1$	$r_p < p < 2r_p$	$0^\circ \leq \nu \leq 360^\circ$	$-90^\circ \leq \phi_{fpa} \leq 90^\circ$
Parabola	$a = \infty$	$e = 1$	$p = 2r_p$	Limited	Limited
Hyperbola	$a < 0$	$e > 1$	$p > 2r_p$	Limited	Limited
Rectilinear Ellipse	$r_p \leq a \leq r_a$	$e = 1$	$p = 0$		
Rectilinear Parabola	$a = \infty$	$e = 1$	$p = 0$		
Rectilinear Hyperbola	$a < 0$	$e = 1$	$p = 0$		

1.3. Two-Body Equation

- Complete knowledge of the two-body problem and its underlying assumptions is crucial to Astrodynamics.
- Newton's second law and his universal law of gravitation are starting points for virtually any study of orbital motion, especially when combined with Kepler's laws.

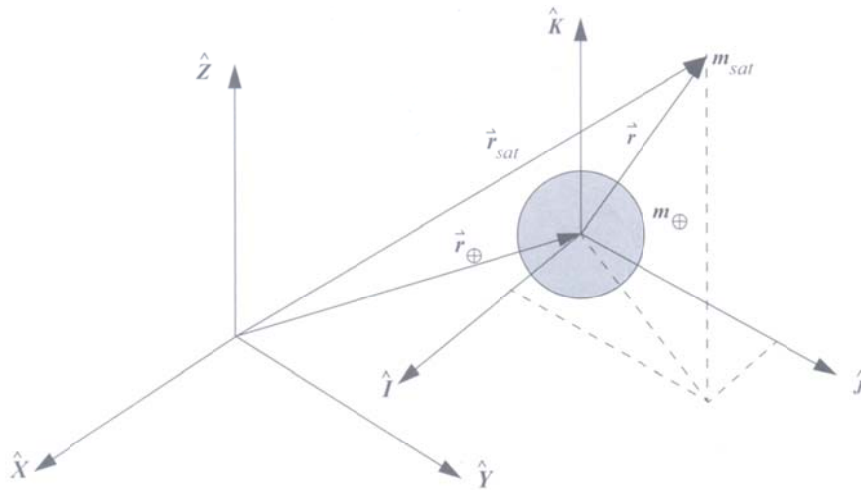


Figure 1-12. Geometry for Two Bodies in an Inertial Reference Frame. XYZ is assumed to be an inertial coordinate system. IJK is displaced from XYZ, but does not rotate or accelerate with respect to XYZ.

- Derivation of Two-Body Equations of Motion (EoM) by applying Newton's Law

$$\vec{F}_{g_{sat}} = m_{sat} \ddot{\vec{r}}_{sat} = -\frac{Gm_{\oplus}m_{sat}}{r^2} \left(\frac{\vec{r}}{r} \right) \Rightarrow \ddot{\vec{r}}_{sat} = -\frac{Gm_{\oplus}}{r^2} \left(\frac{\vec{r}}{r} \right) \quad (1)$$

$$\vec{F}_{g_{\oplus}} = m_{\oplus} \ddot{\vec{r}}_{\oplus} = \frac{Gm_{\oplus}m_{sat}}{r^2} \left(\frac{\vec{r}}{r} \right) \Rightarrow \ddot{\vec{r}}_{\oplus} = \frac{Gm_{sat}}{r^2} \left(\frac{\vec{r}}{r} \right) \quad (2)$$

$$\begin{aligned}
\ddot{\vec{r}} &= \ddot{\vec{r}}_{\oplus sat} = \ddot{\vec{r}}_{sat} - \ddot{\vec{r}}_{\oplus} \\
&= -\frac{Gm_{\oplus}}{r^2} \frac{\vec{r}}{r} - \frac{Gm_{sat}}{r^2} \frac{\vec{r}}{r} \\
\Rightarrow \ddot{\vec{r}} &= -\frac{G(m_{\oplus} + m_{sat})}{r^2} \frac{\vec{r}}{r} \\
\Rightarrow \ddot{\vec{r}} &= -\frac{\mu}{r^2} \frac{\vec{r}}{r}, \quad \mu = G(m_{\oplus} + m_{sat}) \tag{3}
\end{aligned}$$

- (1,2), (1,3) or (2,3) are different representations of Two-Body Equations of Motion
- Need 12 constants for complete knowledge of the Two-Body Problem
 - ◆ Integrals of Motion = Constants of Motion
 - ◆ Initial Conditions are a kind of integrals of motion
 - ◆ Restricted Two-Body Problem
 - Assume $m_{\oplus} \gg m_{sat}$ such that the motion of primary body ($m_{\oplus}$) is not affected by the secondary body (m_{sat})

That is, the primary body is assumed to be fixed in the inertial frame

 - Only 6 constants of motion from the 'restricted' assumption for complete knowledge of the Restricted Two-Body Problem

1.3.1. Assumptions for the Two-Body Equation

- Four Main Assumptions
 - The mass of the satellite is negligible compared to that of the attracting body (Restricted Two-Body Problem)
 - The coordinate system is Inertial
 - ◆ Removes derivatives of the coordinate system itself when differentiating vectors

- ◆ Heliocentric Coordinate System for Interplanetary Probes
- ◆ Geocentric Equatorial System for Earth-orbiting Satellites
- ◆ There does not exist an 'Absolute' Inertial Frame
- Satellite and Attracting body are Spherically Symmetrical with uniform density, and can be treated as a point mass
- No other forces except gravitational forces

1.3.2. Specific Angular Momentum

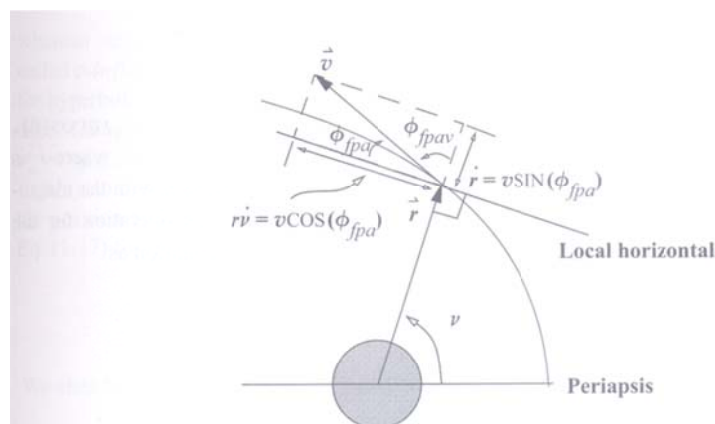


Figure 1-13. Geometry of the Flight-Path Angle. The flight-path angle, ϕ_{fpa} , is measured from the local horizontal to the velocity vector, which we can break into radial and transverse components. The vertical angle ϕ_{fpa_v} is sometimes used in derivations.

- Conservation

$$\begin{aligned}
\vec{h} &= \vec{r} \times \vec{v} = \vec{r} \times \dot{\vec{r}} \\
\Rightarrow \frac{d\vec{h}}{dt} &= \frac{d}{dt}(\vec{r} \times \dot{\vec{r}}) \\
&= \dot{\vec{r}} \times \dot{\vec{r}} + \vec{r} \times \ddot{\vec{r}} \\
&= \vec{r} \times \left(-\frac{\mu}{r^3} \vec{r} \right) \\
&= \vec{0} \\
\Rightarrow \vec{h} &= \text{constant}
\end{aligned}$$

$\Rightarrow$ 3 integrals of motion or 3 partial solutions of two-body problem

➤ Solving differential equations is equivalent to finding integrals of motion

$$\ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r} \Rightarrow \begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ \dot{v}_x = -\frac{\mu}{r^3} x \\ \dot{v}_y = -\frac{\mu}{r^3} y \\ \dot{v}_z = -\frac{\mu}{r^3} z \end{cases} \Rightarrow \begin{bmatrix} x \\ y \\ z \\ v_x \\ v_y \\ v_z \end{bmatrix} = f(x, y, z, v_x, v_y, v_z) \Rightarrow \text{algebraic relation}$$

● Properties

■ $\vec{h} = \vec{r} \times \vec{v}$

$$\Rightarrow h = rv \sin(90 - \phi_{fpa}) = rv \cos \phi_{fpa} \quad (4)$$

$$\Rightarrow h = \begin{cases} r_p v_p & \text{at periapsis} \\ r_a v_a & \text{at apoapsis} \end{cases}$$

In polar coordinates,

$$\vec{r} = r\hat{r}$$

$$\dot{\vec{r}} = \dot{r}\hat{r} + r\dot{\hat{r}}$$

In Fig 1-13,

$$r\dot{v} = v \cos(\phi_{fpa})$$

$$\Rightarrow \cos(\phi_{fpa}) = \frac{r\dot{v}}{v}$$

From (4),

$$h = rv \cos \phi_{fpa}$$

$$= rv \frac{r\dot{v}}{v}$$

$$\Rightarrow h = r^2 \dot{v}$$

■ It turns out that

$$h^2 = \mu p$$

$$\Rightarrow p = \frac{h^2}{\mu} = a(1 - e^2)$$

which relates geometry to dynamics and will be justified in deriving Kepler's 1st law

1.3.3. Specific Energy (Kinetic + Potential)

● Conservation

$$\xi = \frac{v^2}{2} - \frac{\mu}{r} = \text{Kinetic} + \text{Potential}$$

$$\Rightarrow \frac{d\xi}{dt} = \frac{d}{dt} \left(\frac{\vec{v} \cdot \vec{v}}{2} - \frac{\mu}{r} \right)$$

$$= \frac{d}{dt} \left(\frac{\dot{\vec{r}} \cdot \dot{\vec{r}}}{2} - \frac{\mu}{r} \right)$$

$$= \dot{\vec{r}} \cdot \ddot{\vec{r}} + \frac{\mu}{r^2} \dot{r}$$

$$\left(\uparrow \dot{\vec{r}} = \dot{r}\hat{r} + r\dot{v}\hat{v} \right)$$

$$\begin{aligned}
&= (\dot{r}\hat{r} + r\dot{\nu}\hat{\nu}) \cdot \left(-\frac{\mu}{r^3} \vec{r} \right) + \frac{\mu}{r^2} \dot{r} \\
&= (\dot{r}\hat{r} + r\dot{\nu}\hat{\nu}) \cdot \left(-\frac{\mu}{r^2} \hat{r} \right) + \frac{\mu}{r^2} \dot{r} \\
&= -\frac{\mu}{r^2} \dot{r} + \frac{\mu}{r^2} \dot{r} \\
&\equiv 0 \\
\Rightarrow \xi &= \frac{v^2}{2} - \frac{\mu}{r} = \text{constant}
\end{aligned}$$

$\Rightarrow$ 1 Integral of Motion

- Properties

$$\begin{aligned}
\xi &= \frac{v^2}{2} - \frac{\mu}{r} \\
&= \frac{v_p^2}{2} - \frac{\mu}{r_p} \\
&\left(\begin{array}{l} h^2 = \mu p \\ h = r_p v_p \\ \Rightarrow r_p^2 v_p^2 = \mu p \\ \Rightarrow v_p^2 = \frac{\mu p}{r_p^2} \quad (\leftarrow p = a(1-e^2), \quad r_p = a(1-e)) \\ \Rightarrow v_p^2 = \frac{\mu(1+e)}{a(1-e)} \end{array} \right) \\
&= \frac{1}{2} \frac{\mu(1+e)}{a(1-e)} - \frac{\mu}{a(1-e)} \\
&= \frac{\mu(1+e) - 2\mu}{2a(1-e)} \\
&= \frac{-\mu(1-e)}{2a(1-e)}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow \xi &= -\frac{\mu}{2a} \\
\Rightarrow \frac{v^2}{2} - \frac{\mu}{r} &= -\frac{\mu}{2a} \\
\Rightarrow \frac{v^2}{2} &= \frac{\mu}{r} - \frac{\mu}{2a} \\
\Rightarrow v^2 &= \mu \left(\frac{2}{r} - \frac{1}{a} \right) \quad \text{vis-viva equation}
\end{aligned}$$

1.3.4. Kepler's First Law (Trajectory Equation)

- Start from the two-body equations of motion:

$$\begin{aligned}
&\left(\ddot{\vec{r}} + \frac{\mu}{r^3} \vec{r} \right) \times \vec{h} \equiv \vec{0} \\
&\quad \equiv \vec{0} \\
\Rightarrow \ddot{\vec{r}} \times \vec{h} + \frac{\mu}{r^3} (\vec{r} \times \vec{h}) &= 0 \tag{5}
\end{aligned}$$

- For the first term of (5),

$$\ddot{\vec{r}} \times \vec{h} = \ddot{\vec{r}} \times \vec{h} + \dot{\vec{r}} \times \dot{\vec{h}} = \frac{d}{dt} (\dot{\vec{r}} \times \vec{h}) \tag{5}'$$

- For the second term of (5),

$$\begin{aligned}
\frac{\mu}{r^3}(\vec{r} \times \vec{h}) &= \frac{\mu}{r^3} \vec{r} \times (\vec{r} \times \vec{v}) \\
&\left(\uparrow \vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C} \right) \\
&= \frac{\mu}{r^3}((\vec{r} \cdot \vec{v})\vec{r} - (\vec{r} \cdot \vec{r})\vec{v}) \\
&\left(\uparrow \vec{r} \cdot \vec{v} = \vec{r} \cdot \dot{\vec{r}} = r\hat{r} \cdot (\dot{r}\hat{r} + r\dot{\nu}\hat{\nu}) = r\dot{r} \right) \\
&= \frac{\mu}{r^3}(r\dot{r}\vec{r} - r^2\vec{v}) \\
&= \frac{\mu}{r^2}\dot{r}\vec{r} - \frac{\mu}{r}\vec{v} \\
&= -\mu\left(\vec{v}\frac{1}{r} + \vec{r}\frac{-\dot{r}}{r^2}\right) \\
&= -\mu\frac{d}{dt}\left(\frac{\vec{r}}{r}\right)
\end{aligned} \tag{5}''$$

■ Note that LHS of (5) = LHS of (5)' + LHS of (5)''

$$\begin{aligned}
\ddot{\vec{r}} \times \vec{h} + \frac{\mu}{r^3}(\vec{r} \times \vec{h}) &= \vec{0} \\
\Rightarrow \frac{d}{dt}(\dot{\vec{r}} \times \vec{h}) - \mu\frac{d}{dt}\left(\frac{\vec{r}}{r}\right) &= \vec{0} \\
\Rightarrow \frac{d}{dt}\left(\dot{\vec{r}} \times \vec{h} - \mu\frac{\vec{r}}{r}\right) &= \vec{0} \\
\Rightarrow \dot{\vec{r}} \times \vec{h} = \mu\frac{\vec{r}}{r} + \vec{B}
\end{aligned} \tag{6}$$

where $\vec{B}$ is a constant vector defined as eccentricity vector and lies in the orbital plane because RHS is perpendicular to the angular momentum

Now dot product (6) by $\vec{r}$:

$$\begin{aligned}
\vec{r} \cdot (\dot{\vec{r}} \times \vec{h}) &= \vec{r} \cdot \left(\mu\frac{\vec{r}}{r} + \vec{B} \right) \\
\Rightarrow \vec{h} \cdot (\vec{r} \times \dot{\vec{r}}) &= \mu r + rB \cos \nu \\
&\left(\uparrow \equiv \vec{h} \right) \\
\Rightarrow h^2 &= \mu r + rB \cos \nu
\end{aligned}$$

Rearrange for r :

$$\Rightarrow r = \frac{h^2}{\mu + B \cos \nu}$$

$$\Rightarrow r = \frac{\frac{h^2}{\mu}}{1 + \frac{B}{\mu} \cos \nu} \quad (7)$$

Recall the trajectory equation from geometry:

$$r = \frac{p}{1 + e \cos \nu}$$

By comparison with (7), we have

$$p = \frac{h^2}{\mu}, \quad e = \frac{B}{\mu}$$

This equation verifies and extends the Kepler's first law, because it does not restrict the motion to an ellipse

Taking time derivative leads to

$$\dot{r} = \frac{p \dot{\nu} e \sin \nu}{(1 + e \cos \nu)^2}$$

$$\left(\uparrow p = r(1 + e \cos \nu) \right)$$

$$\Rightarrow \dot{r} = \frac{r \dot{\nu} e \sin \nu}{1 + e \cos \nu}$$

1.3.5. Kepler's Second and Third Laws

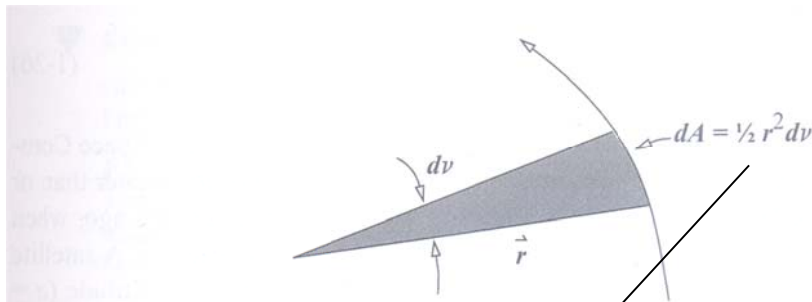


Figure 1-14. Area Swept Out by a Satellite. According to Kepler's second law, satellites sweep out equal areas in equal time intervals.

- Recall (1-18):

$$\begin{aligned}
 h &= r^2 \dot{\nu} \\
 \Rightarrow h &= r^2 \frac{d\nu}{dt} \\
 \Rightarrow dt &= \frac{r^2 d\nu}{h} \quad (*) \\
 \Rightarrow dt &= \frac{2dA}{h} \\
 \Rightarrow \frac{dA}{dt} &= \frac{h}{2} = \text{constant (areal velocity)} = \text{Kepler's Second Law}
 \end{aligned}$$

- Integrate the above relation (*) for one orbital period $[0, P]$, which is equivalent to $[0, 2\pi]$ aerawise:

$$\begin{aligned}
 dt &= \frac{2dA}{h} \\
 \Rightarrow \int_0^P dt &= \int_0^{2\pi} \frac{2}{h} dA \\
 \Rightarrow P &= \frac{2\pi ab}{h} \quad (\because \text{area of ellipse is known to be } \pi ab) \quad (\oplus)
 \end{aligned}$$

Recall the relation:

$$\begin{cases} b = \sqrt{a^2(1-e^2)} = \sqrt{ap} \\ p = \frac{h^2}{\mu} \end{cases} \Rightarrow b = \sqrt{a \frac{h^2}{\mu}} = h \sqrt{\frac{a}{\mu}}$$

which is introduced into $(\oplus)$ to yield

$$P = \frac{2\pi ab}{h} = \frac{2\pi ah \sqrt{\frac{a}{\mu}}}{h} = 2\pi \sqrt{\frac{a^3}{\mu}}$$

$$\Rightarrow P^2 = \frac{4\pi^2}{\mu} a^3$$

Often used to categorize orbits:

- ◆ Low Earth Orbit (LEO)
- ◆ Mid Earth Orbit (MEO)
- ◆ Geosynchronous Orbit (GEO)
- ◆ Geostationary Orbit
- ◆ Semi-Synchronous Orbit
- ◆ Super-Synchronous Orbit

$$\Rightarrow n = \frac{2\pi}{P} = \sqrt{\frac{a^3}{\mu}} \quad \text{mean motion } rad/time(hr, min, sec))$$

$$\Rightarrow n^2 a^3 = \mu \equiv G(m_1 + m_2)$$

$$\Rightarrow a = \left[\mu \left(\frac{P}{2\pi} \right)^2 \right]^{\frac{1}{3}} = \left(\frac{\mu}{n^2} \right)^{\frac{1}{3}}$$

1.3.6.Velocity Formulae

- Start from the definition of Specific Energy:

$$\xi = \frac{v^2}{2} - \frac{\mu}{r}$$

$$\Rightarrow v = \sqrt{2 \left(\frac{\mu}{r} + \xi \right)}$$

$$\left(\Uparrow \xi = -\frac{\mu}{2a} \right)$$

$$\Rightarrow v = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} \quad (*)$$

Recall the trajectory equation:

$$r = \frac{p}{1 + e \cos \nu}$$

$$\Rightarrow r = \frac{a(1 - e^2)}{1 + e \cos \nu}$$

$$\Rightarrow \frac{1}{a} = \frac{1 - e^2}{r(1 + e \cos \nu)} \quad (\oplus)$$

$$\Rightarrow (\oplus) \text{ into } (*)$$

$$\Rightarrow v = \sqrt{\frac{\mu}{r} \left(2 - \frac{1 - e^2}{1 + e \cos \nu} \right)} \quad (\text{General Formula for Ellipse and Hyperbola})$$

For circles ($e = 0$),

$$v_{\text{circle}} = \sqrt{\frac{\mu}{r}} = \text{constant}$$

For parabola which is open and $a \rightarrow \infty$

$$\xi = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}; \quad 0$$

$$\Rightarrow v_{\text{parabola}} = v_{\text{escape}} = \sqrt{\frac{2\mu}{r}}$$

1.4. Three-Body and N -Body Equations

- The standard Two-Body relations are fundamental in Astrodynamics
- Two Body $\Rightarrow$ Three-Body $\Rightarrow N$ -Body

1.4.1. Inertial, Relative and Barycentric Formulas

- Three-Body Problem
 - Again, start from the Newton's Second Law and his law of Gravitation

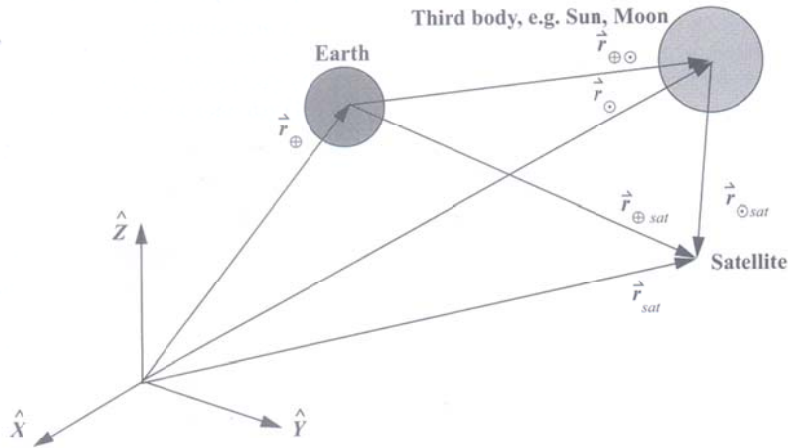


Figure 1-15. Three-body Geometry. The inertial coordinate system XYZ is used to measure all the accelerations. The double subscripts are necessary to completely define the direction of each vector. The satellite is intended to be orbiting the Earth, but for clarity, I've greatly exaggerated the figure.

$$\sum \vec{F}_{g\oplus} = m_{\oplus} \ddot{\vec{r}}_{\oplus} = \frac{Gm_{\oplus}m_{sat}\vec{r}_{\oplus sat}}{r_{\oplus sat}^3} + \frac{Gm_{\oplus}m_e\vec{r}_{\oplus e}}{r_{\oplus e}^3} \quad (1)$$

$$\sum \vec{F}_{gsat} = m_{sat} \ddot{\vec{r}}_{sat} = -\frac{Gm_{\oplus}m_{sat}\vec{r}_{\oplus sat}}{r_{\oplus sat}^3} - \frac{Gm_e m_{sat}\vec{r}_{e sat}}{r_{e sat}^3} \quad (2)$$

$$\sum \vec{F}_{ge} = m_e \ddot{\vec{r}}_e = \frac{Gm_e m_{sat} \vec{r}_{e sat}}{r_{e sat}^3} - \frac{Gm_{\oplus} m_e \vec{r}_{\oplus e}}{r_{\oplus e}^3} \quad (3)$$

■ (1)-(3) constitute a general Three-Body Problem

- ◆ 3 second order vector differential equations
- ◆ 9 second order scalar differential equations
- ◆ 6 first order vector differential equations
- ◆ 18 first order scalar differential equations

■ From (1)-(2),

$$\begin{aligned} \ddot{\vec{r}}_{\oplus} &= \frac{Gm_{sat} \vec{r}_{\oplus sat}}{r_{\oplus sat}^3} + \frac{Gm_e \vec{r}_{\oplus e}}{r_{\oplus e}^3} \\ \ddot{\vec{r}}_{sat} &= -\frac{Gm_{\oplus} \vec{r}_{\oplus sat}}{r_{\oplus sat}^3} - \frac{Gm_e \vec{r}_{e sat}}{r_{e sat}^3} \\ \Rightarrow \ddot{\vec{r}}_{\oplus sat} &= \ddot{\vec{r}}_{sat} - \ddot{\vec{r}}_{\oplus} \\ &= -\frac{Gm_{\oplus} \vec{r}_{\oplus sat}}{r_{\oplus sat}^3} - \frac{Gm_e \vec{r}_{e sat}}{r_{e sat}^3} - \frac{Gm_{sat} \vec{r}_{\oplus sat}}{r_{\oplus sat}^3} - \frac{Gm_e \vec{r}_{\oplus e}}{r_{\oplus e}^3} \end{aligned}$$

■ Using $\ddot{\vec{r}}_{sate} = -\ddot{\vec{r}}_{e sat}$, we obtain 'relative form'

$$\begin{aligned} \ddot{\vec{r}}_{\oplus sat} &= -\frac{G(m_{\oplus} + m_{sat}) \vec{r}_{\oplus sat}}{r_{\oplus sat}^3} + Gm_e \left(\frac{\vec{r}_{sate}}{r_{sate}^3} - \frac{\vec{r}_{\oplus e}}{r_{\oplus e}^3} \right) \\ &= A + B + C \end{aligned} \quad (4)$$

- ◆ A = Two-Body Acceleration of the Earth Action on the Satellite
- ◆ B+C = Perturbation or Additional Forces beyond Two-Body Problem
 - B = Direct Effect = Acceleration of the Third Body directly on the Satellite
 - C = Indirect Effect = Acceleration of the Third Body on the Earth

- Generalization to N-Body Problem

$$\ddot{\vec{r}}_i = -G \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_j}{r_{ji}^3} \vec{r}_{ji}, \quad \vec{r}_{ji} = \vec{r}_i - \vec{r}_j, \quad i = 1, \dots, n$$

- More useful if viewed from the center of mass of the primary body

Derive Analogously to (4), we have

$$\ddot{\vec{r}}_{1sat} = -\frac{G(m_1 + m_{sat})\vec{r}_{1sat}}{r_{1sat}^3} + G \sum_{j=3}^n m_j \left(\frac{\vec{r}_{satj}}{r_{satj}^3} - \frac{\vec{r}_{1j}}{r_{1j}^3} \right)$$

- ◆ 1 = primary body, 2 = satellite

- The equations of motion is not affected by a specific choice of the origin as long as the frame is Inertial and the axis orientation is the same
- Thus, we obtain the same Equations of Motion with respect to the Barycenter (Center of Mass) in the Inertial Frame

$$\ddot{\vec{r}}_{Bsat} = \ddot{\vec{r}}_{sat} = -G \sum_{\substack{j=1 \\ j \neq 3}}^n \frac{m_j \vec{r}_{jsat}}{r_{jsat}^3}$$

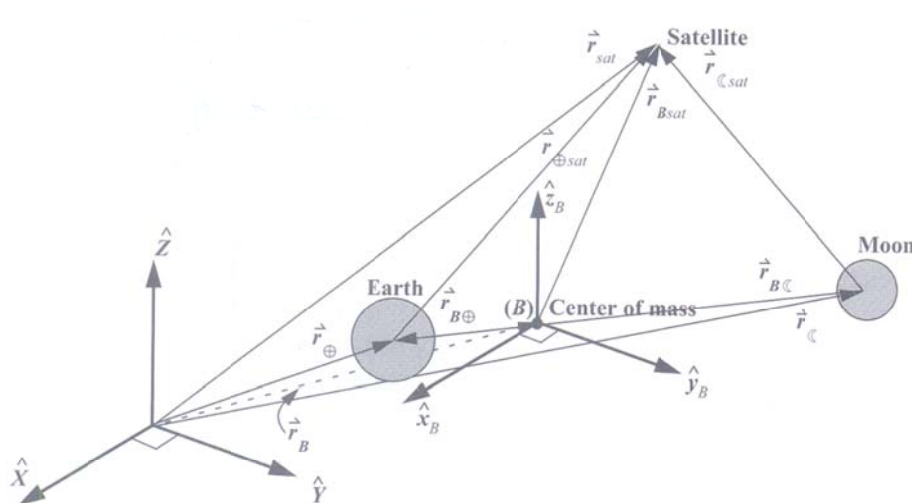


Figure 1-16. Geometry for the Barycentric Equations of Motion. Remember that the barycentric coordinate system is aligned with the inertial XYZ system. The Z and z_B axes are aligned.

- ◆ Not valid for the Restricted Three-Body Problem where the frame is synodic (rotating) with respect to the Barycenter

1.4.2. Ten Known Integrals

- Given the initial position and velocity, we like to solve the equations of motion represented by differential equations
 - Not always able to solve completely
 - Integrals of Motion reduce the order of differential equations or the degree of freedom of the dynamical system
 - The number of Integrals of Motion = the Order of Differential Equations $\Leftrightarrow$ The Problem is solved!

- Two-Body Problem

$$\ddot{\vec{r}}_1 = -\frac{\mu}{r_{21}^3} \vec{r}_{21} \quad , \quad \ddot{\vec{r}}_2 = -\frac{\mu}{r_{12}^3} \vec{r}_{12}$$

- Represented by 2n second order scalar differential equations
- Need 12 Integrals of motion to be completely solved
 - ◆ Conservation of Linear Momentum gives 6 Integrals of Motion

$$m_i \dot{\vec{r}}_i = (\Sigma m_i) \dot{\vec{r}}_{cm} = constant$$

Introducing $\vec{r}_0, \dot{\vec{r}}_0$ provides 6 integrals of motion

- ◆ Specific Angular Momentum provides 3 Integrals of Motion
- ◆ Specific Energy provides 1
- ◆ The eccentricity e is found to be constant from Kepler's law
- ◆ Time of periapsis passage, T , provides the last integral of motion
- * With these 12 integrals of motion, we can say that the Two-Body Problem has been Completely Solved

- General N -Body Problem
 - Represented by $3n$ second order scalar differential equations
 - Need $6n$ integrals of motion to completely solve the problem
 - ◆ Conservation of Linear Momentum gives 6
 - ◆ Conservation of Energy gives 1
 - ◆ Conservation of Angular Momentum gives 3
 - * So far only 10 integrals of motion has been found
 - Thus we are left with a system with $(6n-10)$ order for $n \geq 3$

- Conservation of Linear Momentum

Analogously to the Two-Body Problem,

$$\begin{aligned}
 \sum_{i=1}^n m_i \ddot{\vec{r}}_i &= -G \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}^3} \vec{r}_{ji} = 0 \text{ (No External Force on the System)} \\
 \Rightarrow \sum_{i=1}^n m_i \dot{\vec{r}}_i &= \left(\sum_{i=1}^n m_i \right) \dot{\vec{r}}_{cm} = m_{total} \vec{v}_{sys} \\
 \Rightarrow \sum_{i=1}^n m_i \vec{r}_i &= \left(\sum_{i=1}^n m_i \right) \vec{r}_{cm} = m_{total} (\vec{v}_{sys} t + \vec{r}_{sys})
 \end{aligned}$$

which can be determined from $(\vec{r}_0, \vec{v}_0)$

Thus, 6 integrals of motion can be obtained

- Conservation of Angular Momentum

$$\begin{aligned}
& \left(\sum_{i=1}^n m_i \ddot{\vec{r}}_i + G \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}^3} \vec{r}_{ji} \right) \times \vec{r}_i = \vec{0} \\
& \Leftrightarrow (\quad A \quad + \quad B \quad) \times \vec{r}_i = \vec{0} \\
& \Rightarrow A \times \vec{r}_i + B \times \vec{r}_i = \vec{0} \quad (*)
\end{aligned}$$

Note that

$$\begin{aligned}
A \times \vec{r}_i &= \sum_{i=1}^n m_i \ddot{\vec{r}}_i \times \vec{r}_i \\
&= \sum_{i=1}^n m_i \frac{d}{dt} (\dot{\vec{r}}_i \times \vec{r}_i) \\
&= \frac{d}{dt} \left[\sum_{i=1}^n m_i (\dot{\vec{r}}_i \times \vec{r}_i) \right] \\
&= - \frac{d}{dt} \left[\sum_{i=1}^n m_i (\vec{r}_i \times \dot{\vec{r}}_i) \right] \\
B \times \vec{r}_i &= G \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}^3} \vec{r}_j \times \vec{r}_i \\
&= G \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}^3} (\vec{r}_i - \vec{r}_j) \times \vec{r}_i \\
&= G \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}^3} (-\vec{r}_j \times \vec{r}_i) \\
&= G \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}^3} (\vec{r}_i \times \vec{r}_j) = 0 \text{ (double summation)}
\end{aligned}$$

Introducing the above relation into (*) leads to

$$\begin{aligned}
& - \frac{d}{dt} \left[\sum_{i=1}^n m_i (\vec{r}_i \times \dot{\vec{r}}_i) \right] = \vec{0} \\
& \Rightarrow \vec{h}_{\text{sys}} = \sum_{i=1}^n m_i \vec{r}_i \times \dot{\vec{r}}_i = \text{constant}
\end{aligned}$$

- Conservation of Energy

$$\ddot{\vec{r}}_i = -G \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_j}{r_{ji}^3} \vec{r}_{ji}$$

$$\Rightarrow m_i \ddot{\vec{r}}_i = -G \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}^3} \vec{r}_{ji}$$

For multi-body problem, we define the potential energy as

$$U = G \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}} \quad (\text{analogous to } U = \frac{\mu}{r})$$

Then, the definition of potential energy provides the (conservative) force acting on the i -th body:

$$\vec{F}_i = \frac{\partial U}{\partial \vec{r}_i} = -G \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}^3} \vec{r}_{ji}$$

Then, the equations of motion becomes

$$m_i \ddot{\vec{r}}_i = \frac{\partial U}{\partial \vec{r}_i}$$

$$\Rightarrow m_i \ddot{\vec{r}}_i \cdot \dot{\vec{r}}_i = \frac{\partial U}{\partial \vec{r}_i} \cdot \dot{\vec{r}}_i$$

$$\Rightarrow \sum_{i=1}^n m_i \ddot{\vec{r}}_i \cdot \dot{\vec{r}}_i = \sum_{i=1}^n \frac{\partial U}{\partial \vec{r}_i} \cdot \frac{d\vec{r}_i}{dt}$$

$$\Rightarrow \frac{1}{2} \sum_{i=1}^n m_i \frac{d}{dt} (\dot{\vec{r}}_i \cdot \dot{\vec{r}}_i) = \frac{dU}{dt}$$

$$\Rightarrow \frac{d}{dt} \left(\frac{1}{2} \sum_{i=1}^n m_i v_i^2 - U \right) = 0$$

$$\Rightarrow \xi \equiv \frac{1}{2} \sum_{i=1}^n m_i v_i^2 - G \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ji}} = \text{constant}$$

1.4.3. Three-Body Problem

- Currently, no one has solved the general three-body problem despite the ten known constants of motion
- Effort to analyze the Three-Body Problem (by simplification)
 - Circular Restricted Three-Body Problem (based on two simplification)
 - ◆ Primary and Secondary Bodies move in Circular Orbits with respect to their Barycenter
 - ◆ The mass of the Third Body is negligible compared to that of major bodies

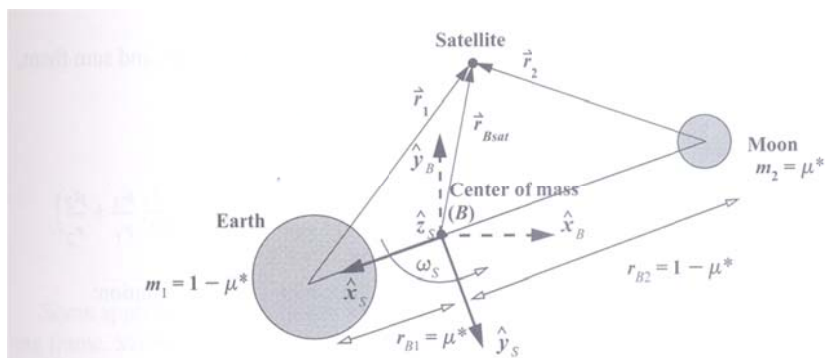


Figure 1-17. Geometry for the Restricted Three-body Problem. Notice the x_S -axis is in the direction of the primary object (Earth), and the y_S -axis lies in the plane of the satellite-Earth-Moon. Remember, this synodic coordinate system is rotating with respect to the fixed barycentric frame (x_B, y_B, z_B). The z_B and z_S axes are aligned. The distances from the barycenter to each object (r_{B1}, r_{B2}) are used to develop the equations of motion. The mass ratio (μ^*) allows us to normalize the problem.

- ◆ Because the frame is not inertial but cynodic, Newton's law is not directly applicable
- ◆ Will study in detail in the Space Flight Dynamics course
- Hill Restricted Three-Body Problem